

ECEN 4/5632 Theory and Application of Digital Filters

Lecture 5, 9-08-26

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* With generous help from Gemini Pro and ChatGPT

The z-Transform and the Region of Convergence

Beyond the DTFT: convergence becomes geometry.

**DTFT = z-transform
on the unit circle**

75 minutes • rigorous DSP foundations • audio examples • Python sandbox • ML bridge

Learning outcomes

00–15 MIN

By the end, students should reason from both algebra and ROC geometry.

- Explain why the DTFT can fail while a z-transform exists elsewhere in the z-plane.
- Compute bilateral z-transforms of common right- and left-sided sequences.
- Use the ROC to distinguish sequences with the same rational expression.
- Perform inverse z-transforms by partial fraction expansion.
- Connect unit-circle inclusion to the existence of the DTFT.

Beyond the DTFT: sometimes the unit circle is the problem

00-15 MIN

The DTFT is only one radial slice of a larger transform.

Example

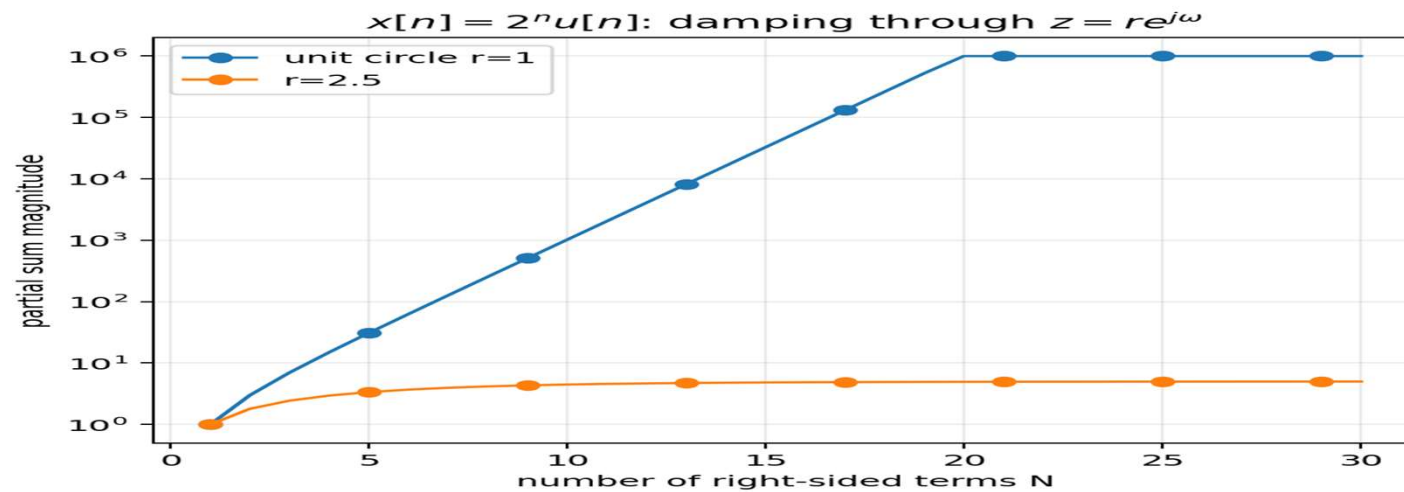
$x[n] = 2^n u[n]$ grows too quickly.

On the unit circle

$\sum 2^n e^{-j\omega n}$ does not converge in the ordinary sense.

Move radially outward

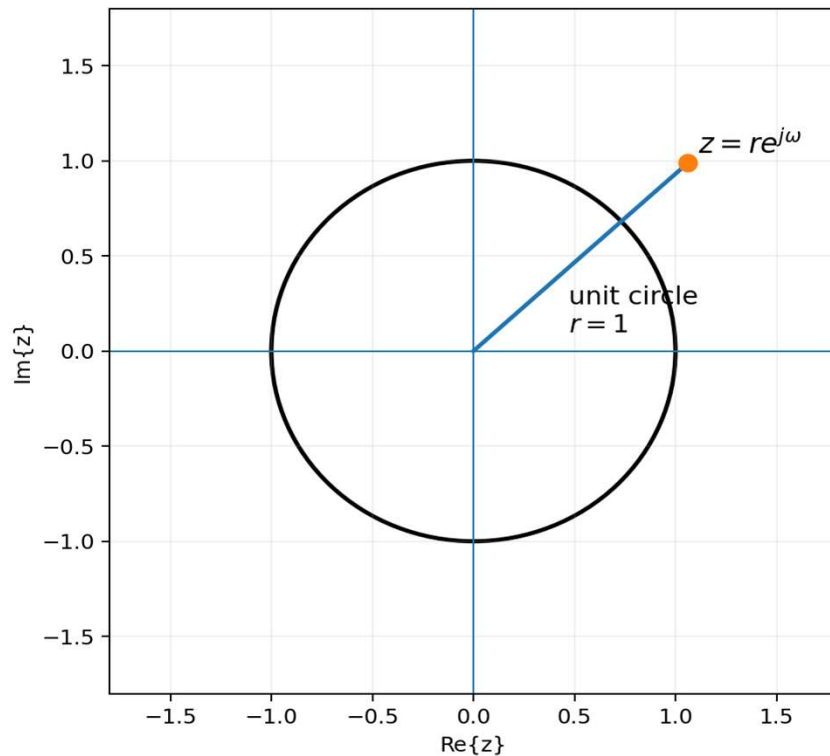
$z^{-n} = r^{-n} e^{-j\omega n}$ adds damping. For $r > 2$, $(2/r)^n$ decays.



The damping-factor interpretation

00-15 MIN

Write z in polar form: radius controls convergence; angle probes frequency.



$$z = r e^{j\omega}$$

- r^{-n} changes the weighting of positive-time samples.
- ω supplies the familiar complex sinusoidal rotation.
- At $r=1$ there is no radial damping: this is exactly the DTFT.
- Convergence can hold for some radii and fail for others.

Definition of the bilateral z-transform

15–35 MIN

A transform value exists only where the infinite sum converges.

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

Complex variable

z can be anywhere in the complex plane—not just on the unit circle.

Transform + ROC

A rational formula without its ROC is incomplete information.

Bilateral

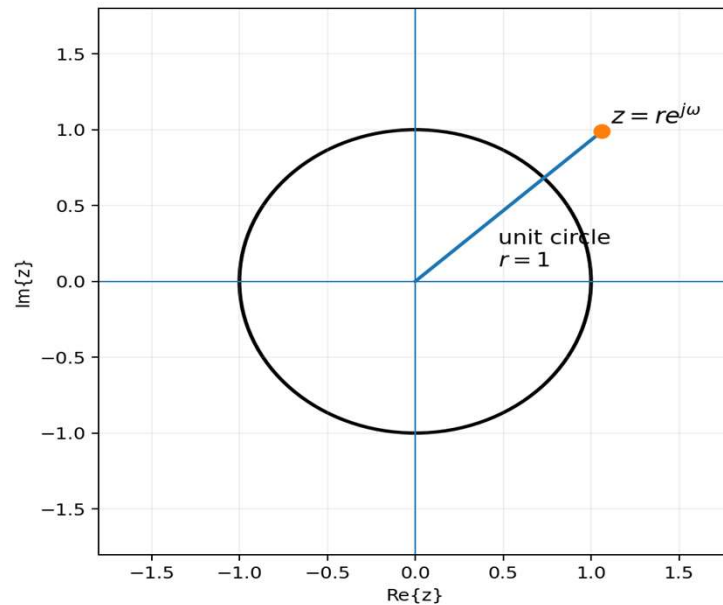
Both positive and negative n may contribute; this is why annular ROCs appear.

The DTFT lives on the unit circle

15–35 MIN

The z-transform generalizes frequency analysis radially.

$$X(e^{j\omega}) = X(z) \text{ at } z = e^{j\omega}$$



- If the ROC includes $|z|=1$, the DTFT exists in the ordinary sense.
- Move around the unit circle: $z=e^{j\omega}$.
- A pole on the unit circle prevents a finite frequency response at that angle.
- Later: stability of an LTI system is tied to the unit circle lying in the ROC.

Standard pair: right-sided exponential

15–35 MIN

Derive it as a geometric series—do not memorize the ROC separately.

$$a^n u[n] \leftrightarrow \frac{1}{1 - az^{-1}}, \quad |z| > |a|$$

Derivation

$$X(z) = \sum_{n=0}^{\infty} a^n z^{-n} = \sum_{n=0}^{\infty} (az^{-1})^n$$

Geometric series converges when $|az^{-1}| < 1$
 $\Rightarrow |z| > |a|$

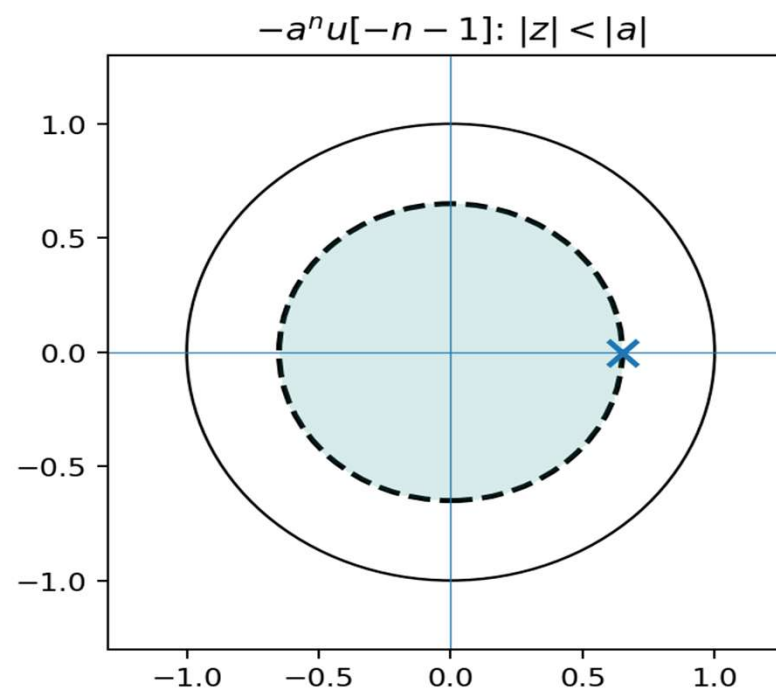
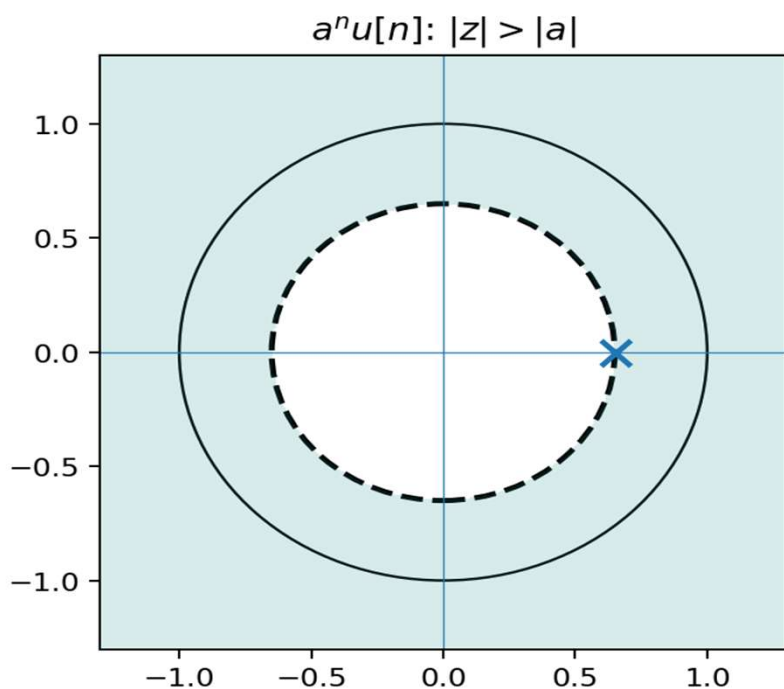
Interpretation

- Pole at $z=a$.
- ROC extends outward from the pole.
- Right-sided sequences naturally produce exterior ROCs.
- If $|a| < 1$, the unit circle lies in the ROC and the DTFT exists.

Same rational expression, different sequence

35-55 MIN

The ROC is what tells you which time support is intended.



Key fact: $1/(1-az^{-1})$ alone does NOT uniquely determine $x[n]$. The ROC is part of the transform.

Left-sided exponential: why the minus sign appears

35–55 MIN

Expand the same rational function in powers appropriate to $|z| < |a|$.

For $|z| < |a|$

$$1/(1 - az^{-1}) = z/(z - a)$$

Rewrite so the geometric ratio has magnitude < 1 :

$$\begin{aligned} &= -(z/a)/(1 - z/a) \\ &= -\sum_{k=1}^{\infty} (z/a)^k \end{aligned}$$

Inverse sequence

$$-a^n u[-n-1]$$

Nonzero only for $n \leq -1$.

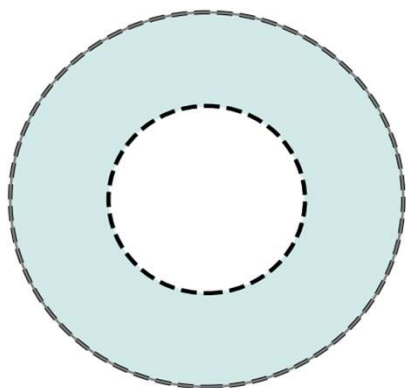
**Same pole at $z=a$.
Different support.
Different ROC.**

ROC geometry: annuli bounded by poles

35–55 MIN

For rational transforms, the ROC cannot contain a pole.

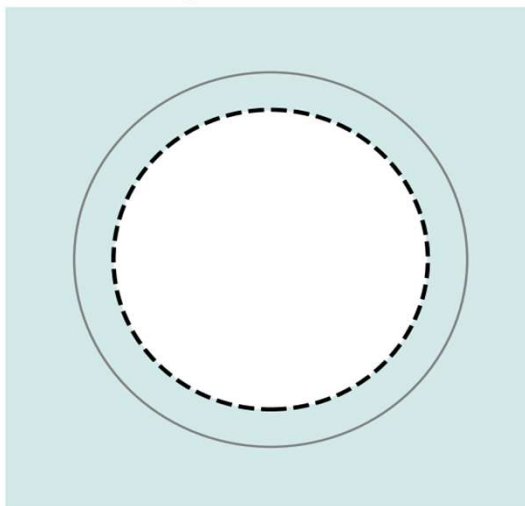
two-sided ROC



Two-sided

ROC lies between pole radii.

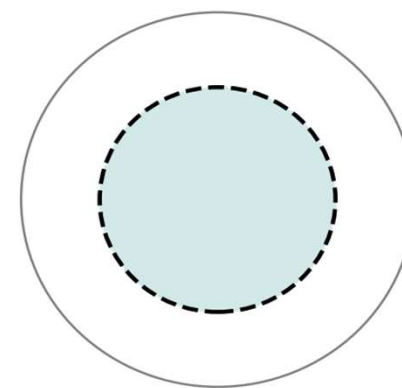
right-sided ROC



Right-sided

ROC is outside the outermost pole.

left-sided ROC



Left-sided

ROC is inside the innermost pole.

ROC rules worth being able to reason from

35–55 MIN

These are consequences of convergence, not arbitrary conventions.

1 ROC is a connected annulus centered at the origin.

2 ROC contains no poles.

3 Right-sided rational sequence $\rightarrow$ ROC outside outermost pole.

4 Left-sided rational sequence $\rightarrow$ ROC inside innermost pole.

5 Finite two-sided sequence: ROC is essentially entire z -plane except possible 0 or ∞ .

6 DTFT exists iff the unit circle lies in the ROC.

Quick classification: infer support from the ROC

35–55 MIN

Think first; compute only if needed.

A $X(z)=1/(1-0.7z^{-1}), |z|>0.7$

B $X(z)=1/(1-1.2z^{-1}), |z|>1.2$

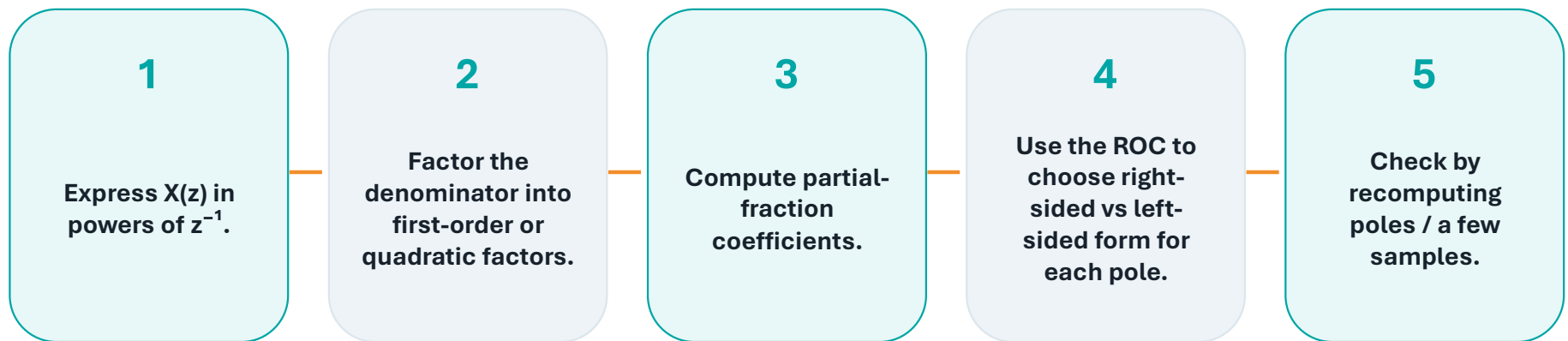
C $X(z)=1/(1-1.2z^{-1}), |z|<1.2$

D poles at 0.5 and 0.9, $0.5<|z|<0.9$

Inverse z-transform: practical workflow

55–75 MIN

For rational functions, partial fractions turns algebra back into transform pairs.



Do not invert the algebraic expression without the ROC: the same residues can map to different time-domain supports.

Worked partial fractions

55–75 MIN

One rational expression; multiple inverse transforms depending on ROC.

$$\begin{aligned} X(z) &= 1 / [(1-0.5z^{-1})(1-0.8z^{-1})] \\ &= -5/3 \cdot 1/(1-0.5z^{-1}) + 8/3 \cdot 1/(1-0.8z^{-1}) \end{aligned}$$

ROC $|z| > 0.8$

Both terms are right-sided.

$$x[n] = -(5/3)(0.5)^n u[n] + (8/3)(0.8)^n u[n]$$

ROC $0.5 < |z| < 0.8$

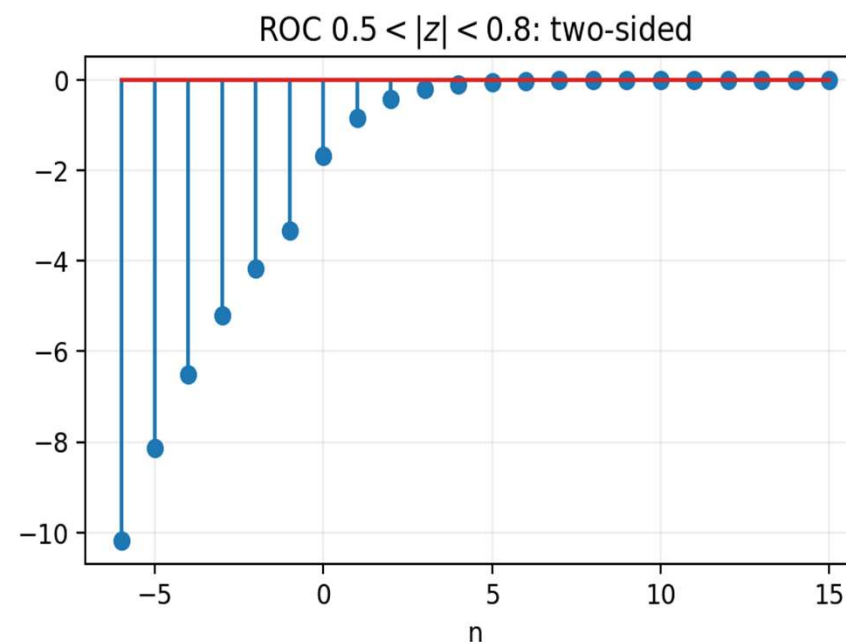
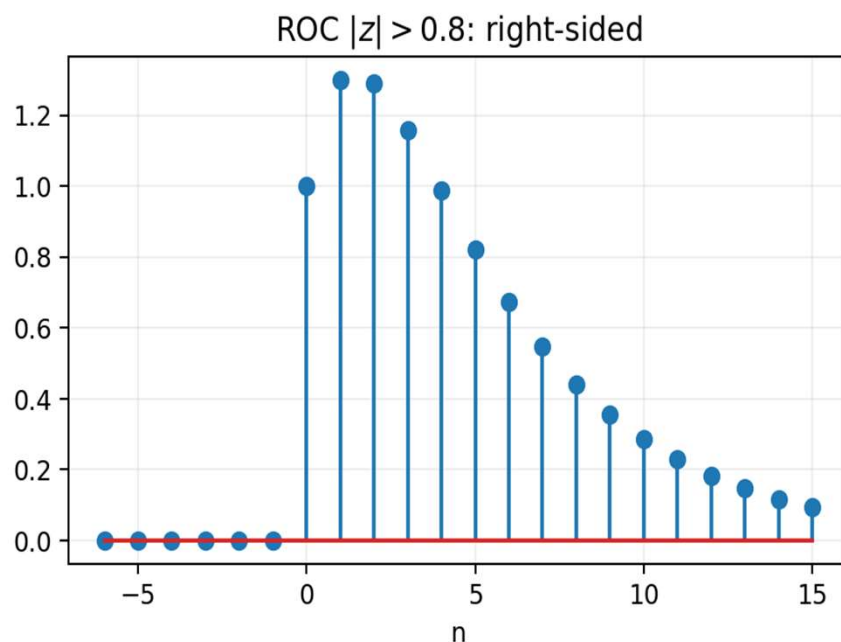
0.5-pole term is right-sided; 0.8-pole term is left-sided.

$$x[n] = -(5/3)(0.5)^n u[n] - (8/3)(0.8)^n u[-n-1]$$

The ROC changes the time-domain signal dramatically

55-75 MIN

Same poles and residues do not imply the same sequence.



This is why “ $X(z)=\text{rational function}$ ” is incomplete unless the ROC (or an equivalent support/causality condition) is specified.

Python demo: see convergence and residues numerically

55–75 MIN

Numerical tools confirm the analysis; they do not decide the ROC for you.

```
from scipy import signal

b = [1.0]
a = [1.0, -1.3, 0.4]
r, p, k = signal.residuez(b, a)
print(r, p)

# residues  $\approx [-5/3, 8/3]$ 
# poles = [0.5, 0.8]
```

What SciPy knows

- Algebraic residues and pole locations.
- A rational transfer-function representation.
- Frequency response if a stable interpretation is appropriate.

What SciPy does NOT infer

- Whether each term is right- or left-sided without an ROC/support condition.

Audio connection: poles forecast decay and resonance

55–75 MIN

Lecture 6 will turn ROC geometry into filter behavior.

Pole magnitude $|p|$

Sets exponential growth/decay rate in impulse-response terms.

Pole angle $\angle p$

Sets oscillatory frequency of a complex-conjugate pole pair.

Unit circle

Frequency response is evaluated here—if this circle lies in the ROC.

Preview: moving a pole closer to the unit circle makes the associated resonance sharper and the time-domain ringing last longer.

Concept check

CHECK

The expression and ROC must be interpreted together.

For $X(z)=1/(1-1.1z^{-1})$, which ROC gives a DTFT?

A. $|z|>1.1$

B. $|z|<1.1$

C. Both

D. Neither

Common mistakes to prevent

WRAP-UP

These are high-value exam and debugging traps.

“The rational formula determines $x[n]$.”

“The z-transform always exists.”

“DTFT and z-transform are separate ideas.”

“A pole can lie in the ROC.”

Lecture 5 synthesis

WRAP-UP

The z-transform is frequency analysis plus convergence geometry.

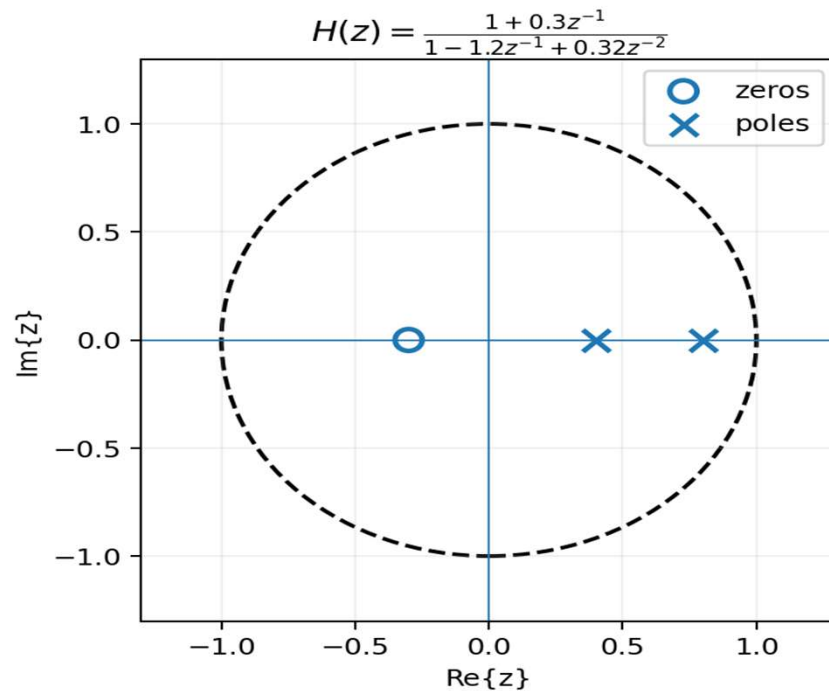


Exit ticket: Why can the same rational $X(z)$ correspond to different sequences?

Next: transfer functions, poles, zeros, stability

NEXT

Lecture 6 turns the z-plane into a filter-design map.



- Difference equations become rational $H(z)$.
- Poles + ROC determine causality and stability.
- Zeros suppress nearby unit-circle frequencies.
- Poles enhance nearby frequencies.
- Audio filters become geometry you can see.